Project Initialization and Planning Phase

Field	Details
Date	05 July 2026
Team ID	(Mee Team ID)
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

The proposed solution aims to improve agricultural productivity using Machine Learning by recommending the most suitable crop based on soil nutrients and environmental conditions. The system predicts the best crop using a trained ML model, helping farmers make informed decisions, increase crop yield, reduce resource wastage, and promote sustainable farming.

Project Overview

Section	Details
Objective	**To develop a Machine Learning based crop recommendation system that suggests the most suitable crop using soil nutrients and environmental conditions.**
Scope	**The project analyzes soil parameters such as Nitrogen, Phosphorus, Potassium, pH, rainfall, humidity, and temperature to recommend the best crop through a Flask web application.**

Problem Statement

Section	Details
Description	Farmers often struggle to choose the right crop due to lack of proper knowledge about soil nutrients and changing environmental conditions. This results in poor crop yield and inefficient use of resources.
Impact	Incorrect crop selection leads to lower agricultural productivity, financial losses, excessive fertilizer usage, and reduced farming efficiency.

Proposed Solution

Section	Details
Approach	Build a Machine Learning model using crop recommendation dataset and deploy it as a Flask web application where users enter soil and weather parameters to receive the best crop recommendation instantly.
Key Features	• Machine Learning based crop prediction• Soil nutrient analysis (N, P, K)• Environmental parameter analysis (Temperature, Humidity, pH, Rainfall)• Fast and accurate crop recommendation• User-friendly Flask web interface

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU specifications	Intel Core i5/i7 Processor (or equivalent)
Memory	RAM specifications	8 GB RAM
Storage	Disk space for project files, models and datasets	20 GB Free SSD/HDD
Software		
Frameworks	Python framework	Flask
Libraries	Machine Learning & Data Processing Libraries	scikit-learn, pandas, numpy, matplotlib, seaborn

Resource Type	**Description**	**Specification / Allocation**
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Dataset	Source, size and format	Kaggle Crop Recommendation Dataset, CSV Format (Soil nutrients, Temperature, Humidity, pH, Rainfall, Crop Label)